



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics and Telecommunication Engineering				
Subject: Engineering Mathematics - III	Code:3000C01AT014				
Maximum/Minimum-Passing Marks: 100/35 Teacher Assessment Marks: 20 Class Test Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	02	01	-	15	45
Class Tests: 2 (Mandatory)	Assignments: 2 (Minimum)				
Duration (ESE): 3 hours	Total Credits: 3				

Course Objectives	Course Outcomes
The objective of this course is to familiarize the prospective engineers 1. To make the students understand that Fourier series analysis is a powerful method where the formulas are integrals and to have knowledge of expanding periodic functions that explore a variety of applications of Fourier series. 2. To provide knowledge of Laplace transform of elementary functions including its properties and applications to solve ordinary differential equations. 3. To have a thorough knowledge of PDE which arise in mathematical descriptions of situations in engineering. 4. To provide a sound background of complex analysis to perform a thorough investigation of major theorems of complex analysis and to apply these ideas to a wide range of problems that include the evaluation of both complex line integrals and real integrals. 5. To study about a quantity that may take any of a given range of values that can't be predicted exactly but can be described in terms of their probability	Course Outcome: After studying the contents of the syllabus in detail the students will be able to 1. Define Fourier series including half range series, Harmonic analysis and variety of its applications. 2. Define (mathematically) Unit step, Unit impulse, Laplace transforms, its properties, Inverse and applications to solve ordinary differential equations. 3. Form and solve by direct integration method Linear equation of first order including Homogeneous and Non-homogeneous Linear equations and also method of separation of variables. 4. Solve difficult problems using theorems of complex analysis and apply Residue theorem to evaluate real integrals. 5. Understand discrete and continuous probability distribution and be able to find mean and standard deviation and use the Uniform distribution.
UNIT-I: FOURIER SERIES Euler's formula, Functions having point of discontinuity, Change of interval, Odd and Even function, Half range series, Harmonic analysis. [9 Hrs] UNIT-II: LAPLACE TRANSFORM Definition, Transform of elementary functions, Properties of Laplace transform, Transform of Derivatives &	



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integrals, Multiplication by t^n Division by t , Evaluation of integrals, Inverse Laplace Transform, Convolution theorem (Without Proof), Unit Step function, Unit impulse function, Periodic function, Application to the solution of ordinary differential equations. [9 Hrs]

UNIT– III: PARTIAL DIFFERENTIAL EQUATION

Formation, Solution by direct integration method, Linear equation of first order, Homogeneous linear equation with constant coefficients, Non-homogeneous linear equations, Method of separation of variables. [9 Hrs]

UNIT–IV: THEORY OF COMPLEX VARIABLES:

Limit, Derivative and Analytic functions; Cauchy-Riemann equations (Cartesian form) Harmonic functions, Flow problems, Complex Integration: Line and Contour integral, Cauchy integral theorem and Integral formula; Taylor series, Laurent series; singularities; Poles and their orders and residues; Evaluation of real definite integrals(Unit Circle).

[9 Hrs]

UNIT–V: RANDOM VARIABLES

Discrete and continuous probability distributions, Expectation, Mean, Standard Deviation & Variance, probability distributions- Binomial, Poisson and Normal distributions.

[9 Hrs]

Text Books:

S. No.	Title	Authors	Edition	Publisher
1)	Higher Engineering Mathematics	B. S. Grewal	45 th Edition, 2024-25	Khanna Publishers
2)	Advanced Engineering Mathematics	H.K. Dass	28 th Edition, 2012	S. Chand Publication
3)	Advanced Engineering Mathematics	Erwin Kreyszig	9 th Edition, 2006	John Wiley & Sons
4)	Applied Engineering Mathematics	Madan Mohan Singh	2 nd Edition, 2016	B S Publications
5)	Engineering Mathematics (Hindi)	S. C. Goyal	6 th Edition, 2022	Khanna Publishers

Reference Books:



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S. No.	Title	Authors	Edition	Publisher
1	Advanced Engg. Mathematics	R.K. Jain and S.R.K. Iyengar	3 rd Edition 2008	Narosa Publishing House
2	Applied Mathematics	P.N.Wartikar & J.N. Wartikar	Vol- II	Vol- II– Pune Vidyarthi Grih Prakashan,Pune
3	Applied Mathematics for Engineers & Physicists	Louis A. Pipes, Lawrence R Harvill	3 rd Edition	Tata Mc Graw-Hill
4	Applied engineering Mathematics-III	Pathak, H.K.	2nd Edition	Shiksha Sahitya Prakashan
5	Calculus and Analytic geometry	G.B. Thomas and R.L. Finney	9th Edition 2002	Pearson, Reprint
6	Higher Engineering Mathematics	B.V. Raman	11 th Reprint 2010	Tata Mc Graw Hill New Delhi
7	A text book of Engineering Mathematics	N.P. Bali and Manish Goyal	Reprint, 2010.	Laxmi Publications
8	Ganit aur Uske Anuprayog	V. K. Singh	5 th Edition, 2021	Arihant Publication

5. **E- resources/ Web Sources:** NPTEL You Tube Channel

6. SWAYAM Portal: Online Lectures on Engineering Mathematics. (https://onlinecourses.swayam2.ac.in/cec20_ma02/preview).



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Electronic Devices and Circuits	Code: 3028C02BT028				
Maximum/Minimum-Passing Marks: 100/35 Teacher Assessment Marks: 20 Class Test Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	2	1	-	15	45
Class Tests: 2 (Mandatory)	Assignments: 2 (Minimum)				
Duration (ESE): 3 hours	Total Credits:3				

Course Objectives:

1. To study various applications of Diode circuits .
2. To understand BJT operation, Biasing techniques and Thermal Stabilization.
3. To provide a thorough understanding of JFET and MOSFET .
4. To familiarize CMOS circuits.
5. To understand special semiconductor devices.

UNIT- I: Applications of Diode Circuits: Rectifier Circuits: Important Characteristics of a Rectifier circuit-Rectifier Efficiency, Ripple Factor, Peak Inverse Voltage, Transformer Utilization Factor etc. Half Wave Rectifier and Full wave rectifier, Need of Filter in Rectifiers. **Zener Diode :** Breakdown mechanism , Characteristics, Voltage Regulator circuit using Zener Diode. **Clippers :** Series Clippers and Parallel Clippers. **Clamper Circuits:** Negative Clamper and Positive Clamper.

UNIT-II: Transistor Biasing and Thermal Stabilization: The DC Load Line and Operating Point, **Stabilization Techniques:** Stability Factor. **Biasing Circuits:** Fixed Bias, Collector to Base Bias , Voltage Divider Bias. **Bias Compensation:** Diode Compensation Techniques, Thermistor and Sensistor Compensation. Thermal Runaway .

UNIT-III: Field Effect Transistor (FET): JFET : JFET Characteristics and Parameters , **JFET Biasing:** Fixed Bias, Self Bias and Voltage Divider Bias. **MOSFET:** Construction , Drain Characteristics and Transfer Characteristics of Depletion Type and Enhancement Type MOSFET, Body Effect, Sub Threshold Conduction.

UNIT-IV: MOSFET Biasing: Comparison of Depletion Type and Enhancement Type MOSFET, MOSFET as a Switch. MOSFET Gate Protection, Applications of MOSFET. **MOSFET Biasing:** Fixed Bias, Self Bias and Voltage Divider Bias. **Basics of CMOS:** CMOS as an Amplifier, Design parameters of CMOS Amplifier, Applications of CMOS.

UNIT-V: Special Semiconductor Devices: Photo Conductive Cell and its Applications, Photodiodes and its Applications, LED and its Applications , Liquid Crystal Displays and its Types , Phototransistor , Photovoltaic Cell, Optoisolator / Optocoupler , its characteristics and advantages . Comparison between LED and LCDs.

Course outcomes:

At the end of this course students will demonstrate the ability to



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1. Analyze Half wave, Full wave rectifiers and understand the functions of clippers and Clampers
2. Solve problems based on Transistor Biasing methods.
3. Describe and analyze parameters of JFET and MOSFET Biasing.
4. Describe and analyze CMOS circuits.
5. Explain working of Special Semiconductor Devices.

Text Books:

S.No	Title	Author(s)	Publisher
1	Integrated Electronics: Analog & Digital Circuit Systems	Jacob Millman & Halkias	Tata McGraw Hill
2	Electronics Circuit Analysis and Design	Donald A Neamen	Tata McGraw Hill
3	Electronics Devices and Circuits	S Salivahanan & N Suresh Kumar	Tata McGraw Hill

Reference Books:

S.No	Title	Author(s)	Publisher
1	Electronic Devices and Circuit Theory	Robert L. Boylestad, L. Nashelsky & K.L. Kishore	PHI
2	Electronics Devices and Circuits	David A Bell	PHI
3	सेमीकंडक्टर भौतिकी, इलेक्ट्रॉनिक्स उपकरण और सर्किट	वाई एस सिसोदिया	-

E-Resources:

<https://nptel.ac.in/courses/108107586>

<https://nptel.ac.in/courses/108108122>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Network Analysis and Synthesis	Code:3028C03BT028				
Maximum/Minimum-Passing Marks: 100/35 Teacher Assessment Marks: 20 Class Test Marks: 20	Periods/week:				
	L	T	P	Ass. Hour	Teaching Hrs/Sem
	2	1	-	15	45
Class Tests: 2 (Mandatory)	Assignments: 2 (Minimum)				
Duration (ESE): 3 hours	Total Credits: 3				

Course Objectives:

1. Apply circuit analysis methods and network theorems to simplify and solve electrical networks effectively.
2. Evaluate initial conditions in elements and networks to analyze transient behavior and system response.
3. Use Laplace Transform techniques to model, analyze, and synthesize standard signal waveforms in circuits.
4. Analyze two port networks using parameter sets and interconnections for practical system applications.
5. Design realizable one-port networks using Hurwitz polynomials, positive real functions, Foster and Cauer forms.

UNIT 1:

(9 Hours)

Methods of Analysing Circuits & Network Theorems: Node and Mesh Analysis, Source transformation, and duality. Network theorems: Superposition, Reciprocity, Thevenin's, Norton's, Maximum Power Transfer, Compensation and Tellegen's theorem.

UNIT 2:

(9 Hours)

Initial Conditions in Networks: Initial Conditions in Elements, Geometrical Interpretation of Derivatives, A Procedure for Evaluating Initial Conditions, Initial State of a Network.

UNIT 3:

(9 Hours)

The Laplace Transformation: Introduction, The Laplace Transformation, Basic Theorems for the Laplace Transformation, Application of Laplace Transformation Technique in Electric Circuit Analysis.

Transforms of Signal Waveforms: Unit Step and Shifted Unit Step Function, Sine and shifted Sine wave, Ramp and Impulse Functions, Waveform Synthesis.

UNIT 4:

(9 Hours)

Two Port Networks: Relationship of Two-Port Variables, Z & Y Parameter, Transmission and inverse Parameters, The Hybrid Parameters, Relationships between Parameters Sets, Interconnection of Two-Port Networks: Series, Parallel and Cascade connection.

UNIT 5:

(9 Hours)

Elements of Realizability and Synthesis of One-Port Networks: Hurwitz Polynomials, Positive Real Functions, Frequency Response of Reactive One-ports, Synthesis of Reactive One-ports by Foster's Forms, Synthesis of Reactive One-ports by the Cauer Method, Synthesis of R-L Network by the Foster Method, Synthesis of R-L Network by Cauer Method, Synthesis of R-C Network by Foster Forms, Synthesis of R-C Network by Cauer Method.

Course outcomes:

After completing this course, students will be able to:

1. Apply node and mesh analysis techniques and network theorems to solve electrical circuits.
2. Determine initial and final conditions in networks and solve transient responses.
3. Utilize Laplace transforms for analyzing circuits in the s-domain.



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4. Model and analyze two-port networks using various parameter sets.
5. Synthesize passive networks using Foster and Cauer forms for given immittance functions

Text Books:

S.No.	Title	Author(s)	Publisher
1	Network Analysis	M. E. Van Valkenburg	Pearson
2	Circuits and Networks Analysis and Synthesis	A Sudhakar , Shyammohan S Palli	McGraw Hill
3	Networks and Systems,	D. Roy Choudhury	New Age International
4	Passive Network Synthesis	M. E. Van Valkenburg	Pearson

Reference Books:

S.No.	Title	Author(s)	Publisher
1	Fundamentals of Electric Circuits,	Charles K. Alexander	McGraw Hill
2	Circuit Theory: Analysis & Synthesis	A. Chakrabarti	Dhanpat Rai & Co.
3	Network Synthesis	B. D. Rao	Pearson Education
4	Circuits and Networks Analysis, Design and Synthesis	M.S. Sukhija, T.K. Nagsarkar	Oxford University Press
5	Network Analysis	G. K. Mithal	Khanna Publishers
6.	नेटवर्क विश्लेषण	तनुज मंगलानी	

E-resources/Web Sources:

<https://nptel.ac.in/courses/108105159>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Digital Logic Circuits	Code:3028C04CT028				
Maximum/Minimum-Passing Marks: 100/35 Teacher Assessment Marks: 20 Class Test Marks: 20	Periods/week:				
	L	T	P	Ass. Hour	Teaching Hrs/Sem
	2	1	-	15	45
Class Tests: 2 (Mandatory)	Assignments: 2 (Minimum)				
Duration (ESE): 3 hours	Total Credits: 3				

Course Objectives:

1. To impart the knowledge of different combinational logic circuits.
2. To understand the basics of Sequential logic circuits.
3. To design, analyse and interpret Sequential Circuits to design counters.
4. To recall synchronous counter types and memory categories, interpret FSM models.
5. To study and remember IC characteristics and logic families.

UNIT I

(9 Hours)

Combinational circuits: Full Adders and Subtractor, Binary Parallel Adder, Look Ahead Carry Adder, BCD Adder, Code Converters, Parity bit Generator and Checker, Encoders and Decoders, Multiplexers and De-multiplexers. Programmable Logic Array, Programmable Array Logic.

UNIT II

(9 Hours)

Sequential Circuits: Latches and Flip Flops: S-R, J-K, Master-Slave J-K Flip-Flop, D-Type and T-Type, Excitation Table of Flip-Flops, Race Around Condition, Flip-Flops, Conversions of Flip-Flops, Application of Flip-Flops, Serial Adder.

UNIT III

(9 Hours)

Shift Registers & Asynchronous Counters: Shift Registers: SISO, SIPO, PISO, PIPO, Bidirectional and Universal Shift Registers. **Asynchronous Counters:** Ripple Up Counter, Down Counter, Up-Down Counter, Design of Mod-N counter (Mod-6, Mod-10 etc.).

UNIT IV

(9 Hours)

Synchronous Counters and Memory Devices: Up counter, Down counter, Up-Down counter, **Design of Synchronous:** Mod-N counter, BCD Counter, Ring Counter, Twisted Ring Counter. **Finite State Machines:** State Diagram, State Table, ASM chart of Mealy and Moore model. **Introduction to Memory Devices:** ROM, RAM, EPROM, EEROM, NVRAM, Flash RAM.

UNIT V

(9 Hours)

Digital Logic Families:

IC Characteristics: Noise margin, Propagation Delay, Fan-in, Fan-out, Voltage and Current Parameters etc. **Logic Families:** TTL Logic, Open Collector, Totem Pole, Tristate TTL, ECL Logic. **MOS Logic:** NMOS Inverter, **CMOS Logic:** Inverter, NAND & NOR Gate. Comparison among various Logic Families.

Course Outcomes: On successful completion of the course, the student will be able to:

CO1: Analyse and Design the Combinational circuits.

CO2: Categorize the Sequential circuits and design different types of Flip-Flops.

CO3: Design and analyze Shift Registers, Asynchronous Counters and Memory devices.

CO4: Design the Synchronous Counter and Finite State Machine.

CO5: Classify the Logic Families based on their performance parameters.

Text Books:



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S.No.	Title	Author(s)	Publisher
1	Fundamentals of Digital Circuits	A. Anand Kumar	PHI
2	Modern Digital Electronics	R.P. Jain	Tata McGraw Hill
3	Digital Fundamentals	Floyd & Jain	Pearson Education
4	Digital Integrated Electronics	H.Taub & Donald Schilling	Tata McGraw Hill.

Reference Books: Title Author(s)\Publisher:

S.No.	Title	Author(s)	Publisher
1	Digital Principles and Applications	Donald P Leach, Albert Paul, Albert Paul Malvino, Goutam Saha	Tata McGraw Hill
2	Digital Circuits and Systems	D.V. Hall	Tata McGraw Hill
3	Digital Electronics	A.P. Malvino	Tata McGraw Hill
4	डिजिटल इलेक्ट्रॉनिक्स	के. के. नेलीवाल	Manju Publication

E-resources/Web Sources:

<https://nptel.ac.in/courses/108105132>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics and Telecommunication Engineering				
Subject: Concept of C++	Code:3028C05CT022				
Maximum/Minimum-Passing Marks: 100/35 Teacher Assessment Marks: 20 Class Test Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	2	1	-	15	45
Class Tests: 2 (Mandatory)	Assignments: 2 (Minimum)				
Duration (ESE): 3 Hours	Total Credits: 3				

Course Objectives:

1. To impart the basic concepts of Object Oriented Programming techniques and C++ programs.
2. To apply the features of object oriented programming to write, compile, execute and debug a C++ programs.
3. To impart knowledge of C++ language.

UNIT-I: Principles of Object Oriented Programming:

(9 Hours)

Basic Concepts of Object Oriented Programming, Benefits of OOPs: Abstraction, Encapsulation, Data Hiding, Inheritance, Overloading, Polymorphism. Classes and Objects: Specifying a Class, Defining Member Functions, Inline Functions, making an Outside Function Inline, Nesting of Member Functions, Private Member Functions, Arrays within a Class, Memory Allocation for Objects, Static Data Members, Static Member Functions, Arrays of Objects, Objects as Function Arguments.

UNIT-II: Friend functions and concepts of Constructor and Destructor:

(9 Hours)

Need of Friendly Functions, Functionality of Friend Function, Constructors and Destructors: Constructors, Default constructors, Parameterized Constructors, Copy Constructors, Multiple Constructors in a Class, Constructors with Default Arguments, Dynamic Initialization of Objects, Destructors and their usage.

UNIT – III: Operator Overloading and Inheritance :

(9 Hours)

Defining Operator Overloading, Overloading Unary Operators, Overloading Binary Operators, Overloading Binary Operators Using Friends, Manipulation of Strings Using Operators, Rules for Overloading Operators, Function Overloading, Defining Derived Classes, Single Inheritance, Making a Private Member Inheritable, Multilevel Inheritance, Multiple Inheritance, Hierarchical inheritance, Virtual Base Classes, Abstract Classes.

UNIT – IV: Pointers and Runtime Binding:

(9 Hours)

Pointers and their Binding, Address Operator & Pointer Variables, Void Pointers, Pointer Arithmetic, Runtime Memory Management, Pointers to Pointers, This Pointer, Introduction to Virtual Functions, Need for Virtual Functions, Pointer to Derived Class Objects, Definition of Virtual Functions, Array of Pointers to Base Class Objects, Pure Virtual Functions, Abstract Classes.

UNIT – V: File Handling:

(9 Hours)

Files: File streams, Hierarchy of file stream classes, Opening and Closing a File, Detecting end-of-file, File Modes, File Pointers and their Manipulations, Error handling during file operations, Reading/writing of files, Accessing records randomly, Updating files. **Templates and Generic Programming:** Template concepts,



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Function templates, Class templates, Illustrative examples. sources, Solution of State equation, Zero state response of the State Vector, Complete Response of the State Vector.

Course Outcomes:

On successful completion of the course, the student will be able to:

1. Describe concepts of objects and classes for developing C++ programs and differentiate between structured and object oriented programming.
2. Develop C++ programs by applying the concepts of friend functions and constructors.
3. Develop C++ programs by applying concepts of operator overloading and inheritance.
4. Apply concepts of Pointers and Runtime binding to develop C++ programs.
5. Use concepts of file handling and exceptions to solve different problems.

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Object Oriented Programming with C++	E. Balaguruswamy	Tata McGraw Hill
2.	Mastering C++	K.R. Venugopal, Raj Kumar and T. Ravi Shankar	McGraw Hill Education
3.	Theory and Problems of Data Structures	Seymour Lipschutz	Schaum's Outline Series

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	C++ Complete Reference Education	H. Schildt	McGraw Hill Education
2.	The C++ Programming Language	Bjarne Stroustrup	Pearson Publications
3	सी++	के.एल. गोदरा	

E-resources/Web Sources:

<https://nptel.ac.in/courses/106105234>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester: 3 rd	Branch: Electronics and Telecommunication Engineering				
Subject: Life Concept by Bhagwat Gita	Code: 3000C06ET046				
Maximum/Minimum-Passing Marks: 100/35 Teacher Assessment Marks: 20 Class Test Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	01	01	-	15	30
Class Tests: 2 (Mandatory)	Assignments: 2 (Minimum)				
Duration (ESE): 3 Hours	Total Credits: 2		Teaching Hours: 30		

Course Objectives:

1. To learn the scientific aspects of the Bhagavad Gita.
2. To explore how the timeless wisdom of the Gita can be integrated into modern life to foster holistic well-being.
3. To discover practical methods from the Gita to manage and reduce stress, anxiety, and overwhelm in personal and professional life.
4. To gain insights from the Gita on building meaningful and harmonious relationships, enhancing effective communication, and critical thinking.
5. To develop mindfulness and meditation practices based on Gita teachings to enhance self-awareness, inner peace, and bhakti for achieving the ultimate goal of life.

Course Outcome:

1. Student understands the basic concepts of Bhagwad-Gita and its need for students.
2. They have been introduced to some concepts of Science of SOULS
3. They have been introduced to some concepts Supreme Lord (Ishwara).
4. They have been introduced to concepts of Yoga – Dhyana, Karma, Jnana, Bhakti

UNIT-I: BASIC CONCEPTS OF BHAGWAD GITA:

(8 Hours)

Introduction to Bhagavad Gita, How to learn Bhagavad-Gita, Five topics in the Bhagavad Gita: Isvara, Jiva, Kala, Karma and Prakriti.

Dilemma in the material world: Arjuna's dilemmas and doubts; perplexities of life in this world: birth, death, disease and old age, three-fold miseries; material relationships: temporary, cause misery; the only solution: spiritual, relationship.

Absolute necessity of spiritual relationship with a bonafide spiritual master/teacher; qualifications of a spiritual master; check and balance through Guru, sadhu and sastra, Qualifications of a disciple/student

UNIT-II: SCIENCE OF SOUL (JIVA):

(6 Hours)

What is soul? Proof of existence of soul, its nature and individuality; the doctrine of life comes from life; activities in knowledge of science of soul. Cause of suffering in this world; constitutional position of living entity.

UNIT-III: SUPREME LORD (ISHWARA):

(8 Hours)

Knowing Krishna, the Supreme personality of Godhead; Krishna as the source of both material and spiritual energies, and the supreme controller; understanding the position of God, His expansions and living entities. Lord's descent: His love, purpose and mission; Paramatma: our ever well-wishing best friend. Inconceivable



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nature of the Supreme Personality of Godhead, His qualities and pastimes; cultivating spiritual relationship: the only way to understand Him. Is God personal or impersonal? Three aspects of the Absolute Truth; path of personalism vs. path of impersonalism. Demigod worship: universal administration; demigods and their relationship with Krishna; right way to approach demigods

UNIT-IV: YOGA (CONNECTING WITH SUPREME LORD):

(8 Hours)

What is yoga? Types of yoga – Dhyana, Karma, Jnana, Bhakti; limitation of asthanga yoga in kali yuga; gradual and direct method - bhakti yoga; pure devotional service; the topmost yoga system. Different platforms of worship: fear, desire, duty,

Karm Yoga (कर्म योग) : Why bad things happen to good people? Law of karma: good karma, bad karma and akarma; how to become free from karma and reincarnation; karma yoga and its importance; contemporary relevance of karma yoga; sakama and nishkama karma yoga

Jnana Yoga (ज्ञान योग): Relation between Karma Yoga and Jñāna yoga, Distinction between Jnana and Vijñāna. Criteria of True Knowledge (Buddhi Yoga & Jñāna Yoga),

Bhakti Yoga भक्ति योग : Four kinds of devotees, Characteristics of Ideal Bhakti- Saraṅāgati & Prapattikrupa (grace); Relation between Bhakti Yoga & Jñāna Yoga

Text Books:

1. Srimad Bhagavad Gita: The ultimate manual of life (1st ed.) Das, M., & Das, T. K. (n.d.) CHOICE (Centre for Heritage of Indian Culture and Education).
2. Introduction to Bhagavad Gita by A.C.Bhaktivedanta Swami Prabhupada.
2. Srimad Bhagavad Gita As It Is. Mumbai: Bhaktivedanta Book Trust, 2009.
3. The Bhagavad Gita. Gorakhpur: Gita Press, 2007.
4. Krishna consciousness - The Topmost Yoga System by A.C.Bhaktivedanta Swami Prabhupada.

Reference Books:

1. Prabhupada, His Divine Grace A.C. Bhaktivedanta. Isopanishad. Mumbai: Bhaktivedanta, Book Trust, 2002.
2. Dasa, Bhurijana. Surrender Unto Me. Vrindavan: Vrindavan Institute of Higher Education, 1997.
3. Prabhupada, His Divine Grace A.C. Bhaktivedanta. The Science of Self-Realization. Mumbai: Bhaktivedanta Book Trust, 2002.

E- Recourses

1. <https://youtu.be/bedsn7xIn0w?si=VW5iQdLgDu5UDhA6>
2. https://anubooks.com/uploads/session_pdf/16623612876.pdf
3. <https://ijcrt.org/papers/IJCRT2101218.pdf>
4. <https://youtu.be/HHIv6qJIRjI?si=MML5aslZRwUvE1Rl>
5. https://en.wikipedia.org/wiki/Jnana-Vijnana_Yoga
6. https://en.wikipedia.org/wiki/Bhakti_yoga



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Electronic Devices and Circuits Lab	Code:3028C01BL028				
Maximum Marks: 40 Teacher Assessment Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	-	-	2	15	30
Duration (ESE): 3 hours	Total Credits:1				

Course Objectives:

1. To provides a platform for students to understand working of active devices such as Diode, BJT, and circuits like rectifier.
2. To construct, test and analyze behavior of FET and MOSFET.
3. Students will become familiar with device characteristics and biasing techniques.

List of Experiments:

1. To Study, implement and perform Series Clipper in Normal and Inverted mode.
2. To Study, implement and perform Parallel Clipper in Normal and Inverted mode.
3. To Study, implement and perform Negative Clamper and Positive Clamper.
4. To Study and Perform Half Wave and Full Wave Rectifier on Breadboard and calculate Ripple factor and Efficiency.
5. To study and verify the working of a full wave bridge rectifier with and without filter and to measure its parameters.
6. To Study and Perform Reverse Zener Diode Characteristic and Zener diode as Voltage regulator.
7. To Build and Test the BJT Fixed Bias and Voltage divider bias circuit.
8. To Study performance of single stage Low Power Common Emitter amplifier.
9. To study behavior of BJT as a Switch.
10. To study Transfer Characteristics of a Junction Field Effect Transistor (JFET).
11. To study the DC behavior and the characteristics of a MOSFET.
12. To design, implement and verify the Self-Bias Circuit operation of FET.
13. To design and verify the Voltage Divider Biasing Circuit of FET.
14. To Study, implement and verify VI Characteristic of LED and find its cut-in voltage.
15. To study and verify the characteristics of, Photodiode and Phototransistor.

Course Outcomes:

After completion of course students will be able to:

1. To construct, test and analyze basic electronic circuits such as rectifiers, filters.
2. To familiarize students with the characteristics of semiconductor devices including diodes, Zener diodes, BJTs, MOSFETs, and FETs through practical experimentation.
3. To design transistor amplifier circuits in different configurations (CE, CC, CB) and analyze their DC performance.
4. To enable students to understand and implement biasing techniques and stability measures for transistors.
5. To Study, implement and perform VI Characteristic of special diodes like LED, Photodiode etc.

Text Books:

S.No.	Title	Author(s)	Publisher
1	Integrated Electronics: Analog & Digital Circuit Systems	Jacob Millman & Halkias	Tata McGraw Hill
2	Electronics Circuit Analysis and	Donald A Neamen	Tata McGraw Hill



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

	Design		
3	Electronics Devices and Circuits	S Salivahana & N Suresh Kumar	Tata McGraw Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1	Electronic Devices and Circuit Theory	Robert L. Boylestad & L. Nashelsky & K. L. Kishore	PHI
2	Electronics Devices and Circuits	David A Bell	PHI

E-Resource:

<https://nptel.ac.in/courses/108108122>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Digital Logic Circuits Lab	Code:3028C02BL028				
Maximum Marks: 40 Teacher Assessment Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	-	-	2	15	30
Duration (ESE): 3 hours	Total Credits:1				

Course Objectives:

1. To verify the Truth Table of different logic circuits.
2. To design and verify the combinational circuits.
3. To design and verify the Sequential circuits.
4. To study RAM, a memory element.
5. To study CMOS and TTL logic gates.

List of Experiments:

1. To design, implement and perform NOR & NAND Gates as Universal building blocks.
2. To design, implement and perform a Full Adder and Full Subtractor Circuit and verify its truth table.
3. To design, implement and perform IC 7483 as a Parallel Adder.
4. To design, implement and perform a 4-Bit Parity Generator and Checker & verify its truth table.
5. To design, implement and perform Binary to Gray, BCD to Excess-3 code converter and vice versa.
6. To design, implement and perform a 4x1 Multiplexer and 1x4 De-Multiplexer using Logic Gates and verify its truth table.
7. To design, implement and perform Octal to Binary Encoder circuit and verify its truth table
8. To design, implement and perform the truth table of multiplexer using 74153 & to verify a demultiplexer using 74139 and design Half-Adder, Half Subtractor, Full Adder and Full Subtractor using multiplexer
9. To design, implement and perform S-R Latch, a J-K, T & D Flip Flop & verify its truth table.
10. To design, implement and perform Serial In Parallel Out, Serial In Serial Out, PISO and PIPO shift registers using IC 7495.
11. To design, implement and perform a 3-Bit Synchronous and Asynchronous Up and Down Counter.
12. To Design Decade Counter, Ring counter/ Johnson counter using IC 7495 and BCD Counter using IC 7490.
13. To store a set of data in RAM using IC 2114.
14. To study IC characteristics, compare and verify the truth table for all logic gates using CMOS logic gates and TTL logic gates.

Course Outcomes:

On successful completion of the course, the student will be able to:

1. Design Full Adder, Full Subtractor, Code Converter and Multiplexers / Demultiplexer.
2. Design S-R, J-K, T and D Flip Flop.
3. Design SISO, SIPO, PISO and PIPO Shift Register.
4. Design and Implement Asynchronous and Synchronous counters.
5. Compare CMOS and TTL logic families.

Text Books:

S. No.	Title	Author(s)	Publisher
1	Fundamentals of Digital Circuits	A. Anand Kumar	PHI
2	Modern Digital Electronics	R.P. Jain	Tata McGraw Hill
3	Digital Fundamentals	Floyd & Jain	Pearson Education
4	Digital Integrated Electronics	H. Taub & Donald Schilling	Tata McGraw Hill.

Reference Books:



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

S.No.	Title	Author(s)	Publisher
1	Digital Principles and Applications	Donald P Leach, Albert Paul, Albert Paul Malvino, Goutam Saha	Tata McGraw Hill
2	Digital Circuits and Systems	D.V. Hall	Tata McGraw Hill
3	Digital Electronics	A. P. Malvino	Tata McGraw Hill

NPTEL Resources:

<https://nptel.ac.in/courses/108105132>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Digital Logic Circuits Lab	Code:3028C02BL028				
Maximum Marks: 40 Teacher Assessment Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	-	-	2	15	30
Duration (ESE): 3 hours		Total Credits:1			

Course Objectives:

- 1.To verify the Truth Table of different logic circuits.
2. To design and verify the combinational circuits.
3. To design and verify the Sequential circuits.
4. To describe RAM as a memory element and compare CMOS logic gates with TTL logic gates.

List of Experiments:

1. To design, implement and perform NOR & NAND Gates as Universal building block.
2. To design, implement and perform a Full Adder and Full Subtractor Circuit and verify its truth table.
3. To design, implement and perform IC 7483 as a Parallel Adder.
4. To design, implement and perform a 4-Bit Parity Generator and Checker & verify its truth table.
5. To design, implement and perform Binary to Gray, BCD to Excess-3 code converter and vice versa.
6. To design, implement and perform a 4x1 Multiplexer and 1x4 De-Multiplexer using Logic Gates and verify its truth table.
7. To design, implement and perform Octal to Binary Encoder circuit and verify its truth table
8. To design, implement and perform the truth table of multiplexer using 74153 & to verify a demultiplexer using 74139 and design Half-Adder, Half Subtractor, Full Adder and Full Subtractor using multiplexer
9. To design, implement and perform S-R Latch, a J-K, T & D Flip Flop & verify its truth table.
10. To design, implement and perform Serial in Parallel Out, Serial In Serial Out, PISO and PIPO shift registers using IC 7495.
11. To design, implement and perform a 3-Bit Synchronous and Asynchronous Up and Down Counter.
12. To Design Decade Counter, Ring counter/ Johnson counter using IC 7495 and BCD Counter using IC 7490.
13. To store a set of data in a RAM using IC 2114.
14. To study IC characteristics, compare and verify the truth table for all logic gates using CMOS logic gates and TTL logic gates.

Course Outcomes:

On successful completion of the course, the student will be able to:

1. Design Full Adder, Full Subtractor, Code Converter and Multiplexers / Demultiplexer.
2. Design S-R, J-K, T and D Flip Flop.
3. Design SISO, SIPO, PISO and PIPO Shift Register.
4. Design and Implement Asynchronous and Synchronous counters.
5. Compare CMOS and TTL logic families.

Text Books:

S. No.	Title	Author(s)	Publisher
1	Fundamentals of Digital Circuits	A. Anand Kumar	PHI
2	Modern Digital Electronics	R.P. Jain	Tata McGraw Hill
3	Digital Fundamentals	Floyd & Jain	Pearson Education



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4	Digital Integrated Electronics	H. Taub & Donald Schilling	Tata McGraw Hill.
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Reference Books:

S.No.	Title	Author(s)	Publisher
1	Digital Principles and Applications	Donald P Leach, Albert Paul, Albert Paul Malvino, Goutam Saha	Tata McGraw Hill
2	Digital Circuits and Systems	D.V. Hall	Tata McGraw Hill
3	Digital Electronics	A. P. Malvino	Tata McGraw Hill

NPTEL Resources:

<https://nptel.ac.in/courses/108105132>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics & Telecommunication Engineering				
Subject: Concept of C++ Lab	Code:3028C03CL022				
Maximum Marks: 40 Teacher Assessment Marks: 20	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	-	-	2	15	30
Duration (ESE): 3 hours	Total Credits:1				

Course Objectives:

1. To apply the concepts of class, method, constructor, instance, data abstraction, function abstraction, inheritance, overriding, overloading, and polymorphism.
2. To analyze, write, debug and test basic C++ codes using the approaches introduced in the course.
3. To analyze problems and implement simple C++ applications using an object oriented software engineering approach.

List of Experiments:

1. Develop C++ program to create a database of students with their Name, Roll no, Marks & Percentage calculations. Print details using classes.
2. Develop C++ program to calculate the sum, difference, product and quotient for two numbers entered by the user as per the user's choice using inline functions.
3. Develop C++ program to find the largest out of two values entered by the user using nesting of Member Functions.
4. Develop C++ program to enter 10 numbers and find the sum & the largest no out of them using Array within the class.
5. Develop C++ program to add time given in hours and minutes using Objects as Function Arguments.
6. Develop C++ program to add two complex numbers using friend function.
7. Develop C++ program to show Overloading of Unary operators and Binary operators.
8. Develop C++ program to show Overloading of Unary operator and Binary operators using friend function.
9. Develop C++ program to explain Overloading of Binary operator for strings manipulations.
10. Develop C++ program to explain the concept of Overloading of functions.
11. Develop C++ program to explain the functionality of Single level Inheritance.
12. Develop C++ program to explain the functionality of Multiple Inheritance.
13. Develop C++ program to explain the functionality of Multilevel Inheritance.
14. Develop C++ program to explain the functionality of Hierarchical Inheritance.
15. Develop C++ program to explain the functionality of Hybrid Inheritance.
16. Implement a C++ program to swap two integers using pointers.
17. Develop a C++ program to reverse a string entered by the user.
18. Implement a C++ program to find the length of a string using concept of pointers.
19. Implement a C++ program to compare two strings using concept of pointers.
20. Implement a C++ program to print and compare volumes of two boxes using classes and pointers.
21. Implement a C++ program to compare ages of three persons using classes and pointers and find out who is the eldest among them.
22. Implement a C++ program to show dynamic memory allocation in C++ using classes and objects.
23. Implement a C++ program to create two arrays to store roll number and marks of some students, whose number would be known at runtime.
24. Implement a C++ program to find the areas of triangle, square and circle using the concept of virtual function.
25. Implement a C++ program to explain the concept of pure virtual function.

Course Outcomes:

On successful completion of the course, the student will be able to:

1. Develop programs for execution of constructor, destructor, static data member, static function and



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member functions.

2. Develop programs to create classes along with constructor and destructor functions.
3. Implement various concepts of classes to concatenate and compare the strings.
4. Develop C++ programs using concepts of Pointers and Runtime Binding to solve practical problems
5. Implement the concepts of file handling and exceptions in C++ programs.

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Object Oriented Programming with C++	E. Balaguruswamy	Tata McGraw Hill
2.	Mastering C++	K.R. Venugopal, Raj Kumar and T. Ravi Shankar	McGraw Hill Education
3.	Theory and Problems of Data Structures	Seymour Lipschutz	Schaum's Outline Series

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	C++ Complete Reference Education	H. Schildt	McGraw Hill Education
2.	The C++ Programming Language	Bjarne Stroustrup	Pearson Publications

E-resources/Web Sources:

<https://nptel.ac.in/courses/106105234>



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester-III	Branch: Electronics and Telecommunication Engineering				
Subject: Idea Innovation Lab	Code:3028C04FL028				
Maximum Marks: 40	Periods/week:				
Teacher Assessment Marks: 20	L	T	P	Ass. Hours	Teaching Hrs/Sem
Duration (ESE): 3 hours	-	-	2	15	30
Total Credits: 1					

Course Objectives

1. To identify a socially relevant or industry-based problem suitable for technological solution development.
2. To analyze user needs and develop a structured solution approach using design thinking.
3. To simulate electronic circuit before hardware implementation.
4. To control electrical appliances using microcontroller and relay module
5. To prepare final project report and presentation for evaluation.

List of Experiments (Perform any 10 experiment from the list, Exp. no. 01 and 15 are compulsory)

1. Identification of Real-World Engineering Problems in Smart Electronics and IoT Domains
2. Application of Human-Centered Design Thinking for Engineering Innovation
3. Systematic Idea Generation Using SCAMPER and TRIZ Innovation Techniques
4. Technical, Economic, and Social Feasibility Analysis of Proposed Smart System
5. Design of Embedded System Architecture Using Block Diagram and Functional Modeling
6. Simulation and Verification of Electronic Circuits Using Industry Standard Tools (Proteus/LTSpice)
7. Sensor Integration and Real-Time Data Acquisition Using Microcontroller Platforms (Arduino/ESP32)
8. Design and Implementation of Actuator Control Systems Using Relay and Power Electronics Interfaces
9. Wireless Communication System Design Using Bluetooth, Wi-Fi, and IoT Communication Protocols
10. Cloud-Based IoT System Development for Real-Time Monitoring and Data Analytics
11. Data Analytics and Visualization of Embedded Sensor Systems Using Python/MATLAB Tools
12. Digital Signal Processing Techniques for Noise Reduction and Sensor Signal Enhancement
13. Prototype Development of Smart Embedded System – Phase I (Hardware Integration and Testing)
14. System Optimization, Debugging, and Performance Enhancement of IoT-Based Prototype – Phase II
15. Technical Documentation, Patent Writing Basics, and Startup-Oriented Project Presentation

Course Outcomes

After successful completion of the course, students will be able to:

1. Identify real-world engineering problems
2. Apply design thinking and innovation methods
3. Develop IoT/embedded systems prototypes and Integrate sensors, controllers, and communication modules
4. Build startup-level engineering solutions
5. Prepare professional technical documentation and presentations

TextBook:

1. Banzi, M. and Shiloh, M., Getting Started with Arduino, 4th Edition, Maker Media, 2022.
2. Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing, 3rd Edition, Springer, 2021.

Reference book:

1. Simon Monk, Programming Arduino: Getting Started with Sketches, 3rd Edition, McGraw-Hill Education, 2022.
2. John R. Karsnitz, Stephen O'Brien, and John P. Hutchinson, Engineering Design: An Introduction, Cengage Learning.



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

3. Autodesk EAGLE Documentation and User Guide.

E-resources/Web Sources:

NPTEL Online Courses

<https://nptel.ac.in> (Courses on Design Thinking, Embedded Systems, Electronics, PCB Design, and Manufacturing.)

<https://swayam.gov.in> (Online courses related to innovation, entrepreneurship, electronics, and design thinking.)



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

Semester: 3 rd	Branch: Electronics & Telecommunication Engineering				
Subject: Personality Development	Code: 3000C07DT076				
Teacher Assessment Marks: 40	Periods/week:				
	L	T	P	Ass. Hours	Teaching Hrs/Sem
	1	-	2	15	30
Duration (ESE): -	Total Credits: -Non Credit Course			Teaching Hours: 30	

Course Objectives: This course aims to develop self-awareness, self-management, communication, interpersonal, and professional skills in students to enhance their personality and prepare them for career and workplace readiness.

Course Contents:

Unit I: **Personality & Self-Awareness** – Concept and determinants of personality, Self-awareness and SWOT analysis, Attitude and confidence building.

Unit II: **Self-Management**– Stress management techniques, Time management and goal setting, Body language.

Unit III: **Interpersonal Skills** – Interpersonal transactions and Transactional Analysis (TA) basics; Conflict – Types and resolution, Digital communication etiquette: email writing, netiquette, and professional social media presence.

Unit IV: **Teamwork** – Group dynamics and team building, Leadership skills, Collaboration and coordination, Problem-solving in teams.

Unit V: **Professional Skills** – Public speaking and presentation skills: overcoming stage fear, structuring a talk, use of visuals; Workplace etiquette: professional appearance, meeting conduct, and workplace relationship.

Course Outcomes:

CO No.	Course Outcome	Bloom's Level
CO1	Identify personality traits and perform self-assessment analysis.	Remember (L1) / Understand (L2)
CO2	Apply self-management techniques for personal and career goals.	Apply (L3)
CO3	Demonstrate effective interpersonal skills in various contexts.	Apply (L3)
CO4	Analyze group dynamics and exhibit teamwork abilities.	Analyze (L4)
CO5	Develop professional competencies and workplace etiquettes.	Create (L6)

Text Books:



CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY BHILAI (C.G.)

S.No.	Title	Authors	Edition	Publisher
1	Personality Development and Soft Skills	Barun K. Mitra	-	Oxford University Press
2	Communication Skills and Personality Development	J.R. Kadam et al.	-	Scientific Publishers

Reference Books:

S.No.	Title	Authors	Edition	Publisher
1	Communication Skills and Personality Development	M. Jegadeesan et al.	-	-
3	How to Win Friends and Influence People	Dale Carnegie	-	-